MCHID Prediction

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Introduction

The goal of this project was to use machine learning to predict MICHD (myocardial infarction or coronary heart disease) based on features extracted from data provided by the Behavioral Risk Factor Surveillance System (BRFSS). In this report we first present our exploration of the dataset, which enabled us to understand its structure and identify potential issues requiring preprocessing. We then describe the strategies developed for handling missing values and standardizing the features, and justify our choices based on performance evaluations. Next, we present the machine learning model trained on the preprocessed data and the methodology used to evaluate its performance. Finally, we discuss the challenges encountered during the project and identify potential improvements.

Data Processing

Analysis of the dataset structure, based on the BRFSS 2015 codebook [2], allowed us to create feature metadata with a semantic grouping of the 321 features (e.g. demographics, cardiovascular, tobacco and alcohol), aiding analysis and interpretability. We recognized that there exist 3 types of missingness: no recorded answer (question not asked, e.g. optional state modules or questions specific to the landline or cell-phone questionnaire), structurally missing answers stemming from question dependencies (e.g. `BPMEDS`, the use of blood pressure medication, is only asked to respondents who reported high blood pressure in `BPHIGH4`), as well as missing answers encoded with non-response codes when the question was asked but no answer was provided (e.g. 7/77 "don't know", 9/99 "refused"). For each feature its non-response pattern was identified. Other non-numeric valid codes were also identified and recoded accordingly (e.g. code 88 "none" in `PHYSHLTH`, the number of days of poor physical health, was recoded to 0, while code 8 "never drive or ride in a car" in `SEATBELT` was kept as a separate indicator outside the ordinal scale).

Another grouping that allowed us to proceed with data imputation and standardization techniques was grouping by data type. The data types that were distinguished: binary, categorical, ordinal, discrete, continuous and encoded, where a single value combines a unit and a quantity (e.g. in `ALCDAY5`, 1xx denotes days per week and 2xx days in the past 30 days). Finally, during the initial pre-processing phase, 25 administrative, date and survey construction features (record identifiers, sampling weights, interview dates) were excluded to not artificially inflate data dimensionality, as they carry no direct information about the respondent's health. The final output of all these operations was a feature metadata dictionary, where for each feature code its semantic category, data type, non-response codes, non-numeric valid codes, value recodes and ordinal levels were available.

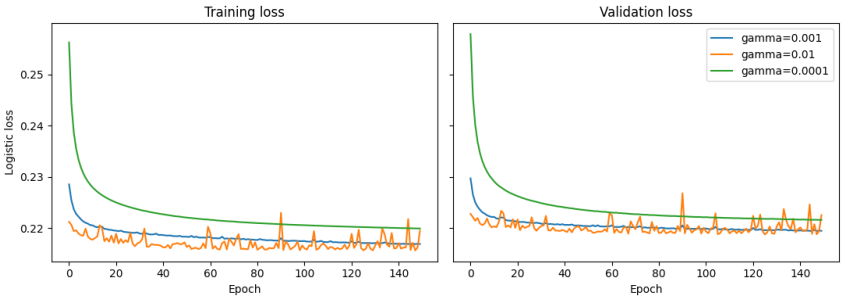
Data Imputation and Standardization

After analysis of the dataset, it became clear that different data types call for different treatments in terms of imputation as well as standardization. It was decided after verifying performance that no imputation would be performed for the binary, ordinal and categorical features, instead the missing values would be encoded. For the continuous and discrete features, we found that different imputation methods (mode, median or random) performed within 0.0013 F1 of this setting in cross-validation, so we kept median for simplicity [1]. Both for imputed and non-imputed data types we explicitly stored

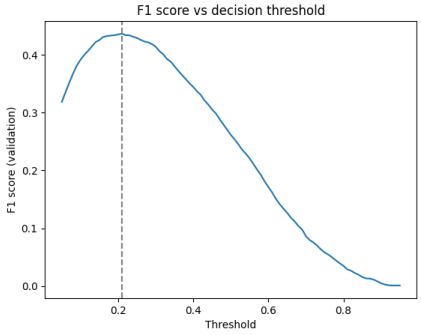
information about the missingness, the empty entries as well as the non-response code were treated as missing. Standardization and missingness were treated differently per data type: binary features were encoded as $-1/+1$, with missing values set to 0, an additional missingness indicator feature was also added; categorical features were one-hot encoded, with missing values represented by an all-zero row; ordinal features were mapped onto equally spaced levels in $[-1, 1]$, a missingness indicator feature and a feature that indicated valid non-numeric codes (e.g. "never drive" in `SEATBELT`) were also added; discrete and continuous features were imputed with the median and standardized, again with additional missingness indicator features. All parameters were learned on the training data only to avoid data leakage.

Optimization Method and Model

After settling on the final dataset (all observations and all features without a missingness threshold, no imputation for binary, categorical and ordinal features, median imputation for discrete and continuous ones), the ridge-regularized logistic regression was optimized with mini-batch SGD (batch size 32). A small convergence study on an 80/20 split over 150 epochs yielded $\gamma = 10^{-3}$ as the best learning rate out of $\{10^{-2}, 10^{-3}, 10^{-4}\}$: $\gamma = 10^{-2}$ reached its plateau quickly but kept oscillating around it, while $\gamma = 10^{-4}$ converged too slowly (see graphs (a)). The number of epochs was chosen to be 100 to balance computational cost and accuracy (validation loss 0.2199 after 100 vs. 0.2195 after 150 epochs). The regularization parameter, as well as the decision threshold (the probability threshold above which someone is declared sick), was then selected using 5-fold CV to maximize the F1 score [3]. It resulted that the best decision threshold was 0.21 (see graph (b)), and that regularization was not needed, as the best F1 score was obtained with $\lambda = 0$.



(a) Training and Validation Losses for different gamma



(b) F1 score vs Decision Threshold

Discussion

Despite satisfactory results, some improvements can still be made. First, our imputation methods only depended on the distributions of each feature individually. However as we observe from the codebook, features can be related in different ways: some features are physically related and, as pointed out before, in some features, missing answers are structural. An improved imputation technique could exploit these properties. An approach we explored was KNN, however it was computationally too expensive as it, for every missing value in a feature, identifies observations that are similar (by using the other features). For that it has to repeatedly go through a large amount of data points which isn't feasible for large datasets. A possible solution would be to apply KNN imputation to groups of features that show apparent physical relations, which would additionally reduce the number of dimensions involved and potentially reduce computational cost. Another improvement could be replacing SGD with Newton's method. Since the loss function of logistic regression is convex, Newton's method, with appropriate step-size control, can converge to the globally optimal weights. Although each iteration is more computationally expensive, its overall faster convergence could reduce the overall computational cost, allowing us to run further experiments without the current upper bound. A last improvement that could be significant is feature augmentation. As our feature analysis showed, some features (e.g. `_AGE80`) have a non-linear relationship with the target. Rather than using a purely linear model, we could add non-linear transformations of these features, which would better capture these effects.

AI acknowledgement

We have used AI (ChatGPT, Claude) for repetitive tasks such as plotting graphs, and for parsing the dataset and the features to create the feature metadata dictionary (see section "Data Processing").

References

[1] Tero Aittokallio. Dealing with missing values in large-scale studies: microarray data imputation and beyond. *Briefings in Bioinformatics*, 11(2):253–264, 2010.

[2] Centers for Disease Control and Prevention. *Behavioral Risk Factor Surveillance System: 2015 Codebook Report*. Centers for Disease Control and Prevention, August 2016.

[3] Zachary Chase Lipton, Charles Elkan, and Balakrishnan Naryanaswamy. Optimal thresholding of classifiers to maximize f1 measure. In *Machine Learning and Knowledge Discovery in Databases*, pages 225–239. Springer, 2014.